Drone-Based Room Occupancy Counter

Project Proposal & Technical Workflow

Objective	Build a low-cost drone that streams its camera feed live to a laptop over Wi-Fi, so the software can detect and count people in real time as the drone flies.
Hardware (proposed)	A basic self-assembled quadcopter frame + an ESP32-CAM module for Wi-Fi video (to be finalized).
Core deliverable	A live video pipeline that reads the drone's Wi-Fi stream on a laptop and reports a real-time person count.

1. Project Goal

The project builds and flies a small, low-cost drone equipped with a Wi-Fi camera, and streams that camera feed live to a laptop, where the software detects and counts the number of people in the room or area in real time. Unlike a record-then-analyze setup, the goal here is for the video to be pulled directly into the code as the drone is flying, so detection and counting happen live rather than after the flight.

The exact hardware (frame size, motors, flight controller, battery) is intentionally left open in this proposal — we would like your guidance on the best components to use within a modest budget. Our own starting suggestion, mainly to keep costs low, is noted below.

2. Our Proposed Hardware Direction

Our starting suggestion is to self-assemble a basic quadcopter frame (motors, ESCs, flight controller, battery, receiver) and mount an ESP32-CAM module as the dedicated video payload. The ESP32-CAM is a small, inexpensive camera board that runs its own tiny web server and broadcasts an MJPEG video stream over its own Wi-Fi access point — the laptop simply connects to that Wi-Fi and reads the stream as a normal video source in code (e.g. via OpenCV), with no phone app or manual transfer step involved.

We're flagging this as a starting point, not a final decision — we'd value your input on frame choice, motor/ESC sizing to carry the extra camera weight, and whether a different video module would serve the project better.

3. End-to-End Workflow

Stage	Task	Output
1. Build	Assemble the drone frame, motors, flight controller and mount the Wi-Fi camera payload; test basic flight stability.	Flight-ready drone
2. Connect	Laptop connects to the camera module's Wi-Fi access point; open its MJPEG stream URL in code.	Live video feed on laptop
3. Fly & Stream	Fly the drone through the target room/area while the video streams continuously to the laptop.	Continuous live frames
4. Detect (live)	Run a person-detection model (e.g. YOLO) on each incoming frame as it arrives.	Per-frame bounding boxes

5. Track (live)	Apply object tracking (e.g. CSRT/KCF or DeepSORT) so the same person keeps one ID across frames.	Unique person IDs, updated live
6. Count	Maintain a running count of unique tracked IDs; display the current headcount on screen during flight.	Live person count
7. Report	Summarize methodology, accuracy checks, and results after the demo flight.	Final report/demo

4. Technical Approach (Software)

4.1 Detection

A pretrained person-detection model (YOLO family) will locate people in each video frame. 720p resolution is sufficient for this task, since the goal is presence/count rather than identification.

4.2 Avoiding Double-Counting

Because the drone moves through the room, the same person appears in many consecutive frames. Counting per-frame detections directly would badly overcount. Two options are being considered:

- Multi-object tracking (assign each detected person a persistent ID across frames, e.g. via DeepSORT or a lightweight tracker such as CSRT/KCF) — count unique IDs.
- A single slow sweep with frame-stitching into one merged panoramic view, counting once from that merged image — simpler, but requires more careful, controlled flight.

4.3 Output

While the drone is flying, the laptop displays the live video with bounding boxes and IDs overlaid, plus a running headcount that updates as new people are tracked. At the end of the flight, this running count becomes the final reported number for the surveyed area.

5. Practical Constraints to Plan Around

- Latency: even with a direct Wi-Fi connection, expect a real delay of roughly a few hundred milliseconds to about a second between what the camera sees and what appears on the laptop — this is "live" in the sense of no manual transfer step, not truly instantaneous.
- Range: a camera module's own Wi-Fi access point typically has a reliable range of only about 30-80 metres indoors, less with walls/interference — fine for a room-scale demo, limiting for larger open areas.
- Payload weight: the video module (plus its antenna and any voltage regulation needed) adds weight the frame/motors must be sized to carry without cutting flight time too short.
- Flight time: small DIY frames at this budget typically give somewhere in the 5-10 minute range per charge — a spare battery is worth planning for.
- Camera resolution is adequate for person detection but not for facial identification.

6. What We Need Help With

- Recommending or approving the specific hardware (frame size, motors/ESCs, flight controller, battery, video module) within our budget.
- Guidance and supervision on safely assembling and flight-testing the drone.
- Feedback on whether live, in-flight detection is a realistic scope, or whether a hybrid (live preview + more thorough analysis afterward) would be more reliable for a first version.
- Any lab space, spare batteries, or safety guidelines for indoor test flights.